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Computer Engineering student with a strong interest in artificial intelligence, data science and software development. Motivated by building practical systems that solve real-world problems, and eager to keep learning across new technical domains.

Experience

ULAK Haberleşme A.Ş.

Ankara, Turkey

Artificial Intelligence Intern

August 2026 – September 2026

- Built ULAK101, a source-grounded enterprise telecom assistant with authorization-aware hybrid retrieval combining BM25 and dense search, RRF fusion, cross-encoder reranking and query rewriting.
- Added Agentic, Corrective and Self-RAG loops over local LLMs, with conversation memory, OCR document processing and AES-256-GCM encrypted storage. Technologies: Python, FastAPI, React, TypeScript, PostgreSQL, ChromaDB, Ollama.

Microsoft

AI Innovators Program

AI Engineer Intern

May 2026 – July 2026

- Delivered a multilingual RAG system with vector search, hybrid BM25/TF-IDF retrieval, cross-encoder reranking and NLLB-200 translation, validated by an automated audit report and Pytest suite. Technologies: Python, Streamlit, ChromaDB, SQLite, Ollama, llama3.1:8b, bge-m3.

TARAFTech GmbH

Germany

Computer Engineering Intern, Full Stack

February 2026 – April 2026

- Designed, developed and deployed production web applications across the full stack.

Nurol Teknoloji

Ankara, Turkey

Project Researcher and Project Manager, innovationNTogether

October 2025 – June 2026

- Delivered an AI-powered demand forecasting system on a 13-table PostgreSQL star schema, training LightGBM with walk-forward cross-validation and Optuna tuning over sales, procurement, lead-time and weather data.
- Fed forecasts into Safety Stock, Reorder Point and EOQ engines producing automatic reorder recommendations; owned sprint planning and shipped FastAPI services, Docker runtime and React/Streamlit dashboards. Technologies: Python, LightGBM, Pandas, Scikit-learn, FastAPI, SQLAlchemy, Docker, Agile.

ETKİM

ODTÜ Teknokent, Ankara

Software Engineer

August 2025 – May 2026

- Developing a decision-driven VR values and career simulator for the Turkish Ministry of National Education, owning scenario design, 3D environments and branching interactions. Technologies: Unity 6.1, OpenXR, C#, XR Interaction Toolkit, PostgreSQL.

Havelsan Teknoloji Radar

Ankara, Turkey

R&D Software Intern, Image Processing

August 2025 – September 2025

- Built an operational 15-camera system re-identifying a target across cameras within 2 seconds without visible faces, recovering targets lost to occlusion through embedding-similarity search.
- Added sabotage detection for camera blocking, laser interference and abnormal tilt with sub-second alarms; demonstrated the working system to HAVELSAN R&D engineers. Technologies: YOLOv11s, TorchReID (OS-Net), BoT-SORT, GStreamer, Flask, WebSocket.

Education

TED University

Ankara, Turkey

B.Sc. Computer Engineering (English). GPA: 3.39. 50% Merit Scholarship.

2022 – Present

Projects

Every project below has a live demo, source code and a full technical report at umut-uygur.vercel.app

- **GPS-Free UAV Position Estimation** (SSB, 2026) — Estimated UAV position during GPS outages from IMU, magnetometer, barometer and airspeed alone, reaching 4.124 m mean horizontal error and 2–3x improvement over dead reckoning. Technologies: PyTorch, GRU, LSTM, AttentionGRU, CNN, Sensor Fusion.

- **Library Occupancy Detection System** (TED University, 2025–2026) — Monitored 2 libraries, 40+ cameras and 400 seats with polygon-based empty/occupied/hold classification, backed by 105,000 TRY of approved institutional funding. Technologies: Python, OpenCV, YOLOv8, Ultralytics, SSIM, NumPy.
- **Portfolio Decision Engine** (KPMG IxT Olympics, 2026) — **Third place among 955 applicants**; macro-data-driven engine recommending dynamic asset allocations by risk profile with automated ingestion and back-testing. Technologies: Python, Pandas, FastAPI, TCMB EVDS, FRED, Selenium, Pytest.
- **UBEB 5G and AI Smart Road Safety** (TEKNOFEST and Turkcell, 2026) — Detected vehicles, license plates and risky in-cabin driver behaviour from dashcam video in one pipeline at 2.5 FPS on 4K input with sub-2 ms JSON output. Technologies: YOLOv8, BoTSORT, OSNet, EasyOCR, MediaPipe, ONNX Runtime, Docker, FastAPI, WebRTC.
- **E-Commerce Query-Product Relevance System** (TEKNOFEST and Trendyol, 2026) — **77th of 377 teams** in a two-person entry, scoring relevance over a 962,000+ item Turkish catalog. Technologies: BERT, Cross-Encoder, Sentence-Transformers, FAISS, BM25, LightGBM, Hard-negative Mining.
- **Hybrid Intelligence for UAV Thermal Imagery** (SAYZEK, 2025–Present) — Analyzing thermal imagery with fine-tuned vision-language models deployed on edge devices in closed networks, with academic publication planned. Technologies: PyTorch, Vision-Language Models, Thermal Image Processing, Fine-Tuning, Linux.
- **Real-World Pile Load Test Data Project** (TED University, 2025) — Modeled stratigraphy and pile capacity from real load-test data with Dr. Cem Akgüner, deriving Chin-Kondner and Decourt targets. Technologies: Pandas, Scikit-learn, Random Forest, LightGBM, CatBoost, SHAP, Optuna.
- **CI/CD Portfolio Deployment Pipeline** (2026) — Automated Docker builds through GitHub Actions to Amazon ECR, deploying to EC2 via Docker Compose with a Spring Boot backend and Vercel frontend. Technologies: GitHub Actions, AWS (ECR, EC2, IAM), Docker, Spring Boot, Java 21, Maven, Linux.
- **Additional projects** — Learning Management System with full relational database (Java Swing, MySQL, JDBC, triggers, role-based access), Softmax Regression Pipeline, MNIST-784 Classifier, K-Means and GMM Image Segmentation, Web Traffic Log Search, Software Development Chatbot, Sorting Algorithm Analysis, Network Programming Assignments.

Leadership & Activities

Kaggle, Master Tier	2025 – Present
Bronze medal in Hull Tactical Market Prediction (238/3,677); datasets and notebooks reaching 157 and 52 upvotes.	
President, TEDU IEEE Student Branch	2024 – 2027
Oversaw student activities and initiatives; Board Member, IEEE CS Committee (2024–2025).	
President, Founder and Instructor, TEDU Partner Dances Society	2022 – 2025
Founded the society, led its activities and instructed members.	
SkillsBuild Student Ambassador, IBM	2026 – 2027
Camp Management and Staff, Herräng Dance Camp, Sweden	2023 – 2025
Worked under pressure in an international team environment with intercultural cooperation.	

Skills & Certificates

Programming: Python, Java, C, C#, SQL, JavaScript, TypeScript, HTML/CSS • **Language:** Turkish (Native), English (Advanced), German (Basic)

Machine Learning: PyTorch, TensorFlow, Scikit-learn, LightGBM, CatBoost, Computer Vision, YOLO, Multi-Object Tracking, Re-Identification, OCR, LLM, RAG, NLP, Time-Series Forecasting, Fine-Tuning, Edge AI

Data & Engineering: Pandas, NumPy, OpenCV, Matplotlib, Seaborn, SHAP, Optuna, PostgreSQL, MySQL, ChromaDB, FAISS, FastAPI, Flask, Spring Boot, React, Streamlit, Docker, GitHub Actions, AWS, Git, Linux, REST APIs, WebSocket, Unity, OpenXR

Certificates: Huawei Cloud Certified Developer Professional (GaussDB(DWS)) • Google Project Management • HKUST Software Engineering Specialization • Milli Teknoloji Akademisi AI Specialist • IBM Generative AI and Data Science Essentials • Data Science and AI Summer School • Huawei Cloud Certified Developer Associate in AI

References

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